



MRIM Installation and Operations Manual

Railway Administration and Information Logical System

RAILS for Model Railroads

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Chapter 1

Model Railroad Information Manager

1.1 Introduction

Railway Administration and Information Logical System (RAILS) is a software model and implementation of an automated system to assist the model railroader achieve realism in the operation of a model railroad. The RAILS system design describes the system's architecture, components, and interfaces and can be found at [RAILS System Design and implementation](#). There are four user interface Single Page Applications (SPAs) that provide different aspects of RAILS they are:

- Rollingstock RFID Manager (RSRM) provides a user with a web application to show, when an RFID tag is read, the road name and number of the rolling stock associated with that tag.
- Model Railroad Inventory Manager (MRIM) the focus of this manual, allows the user to create, update and delete model railroad assets, such as:
 - manage lists of items found in model railroads:
 - ✱ Rolling Stock
 - ✱ DCC decoders
 - ✱ AAR Codes
 - ✱ Images
 - ✱ Companies
 - ✱ Structures
 - creates PDF reports; and
 - ✱ AAR Codes
 - ✱ Companies
 - ✱ Images
 - ✱ RFID
 - ✱ Rolling Stock sortable by Road Name and Road Number or AAR Code or Status
 - exports and imports inventories in CSV files.
- Model Projects and Purchase Manager (MPPM) allows the user to enter information about their projects and purchases; and
- Model Railroad Layout Manager (MRLM) allows the user to enter information about their layout and control elements of it.

1.2 Components

The implementation of MRIM consists of the following microservices components:

- Model Railroad File Manager (MRFM) provides the user the ability to upload image files;
- images is a repository for the images of rollingstock and structures;
- Railroad Inventory Data Services (RIDS) provides Representational State Transfer (REST) access to railroad inventory collections including rollingstock (RS);
- MongoDB a NoSQL database program that stores data records as documents which are gathered in collections. A database stores one or more collections of documents;
- Model Railroad (MR) Data is the document repository, used by MongoDB, to store complete collections of items such as RS, industries (producers and consumers), track elements, turnouts, projects, purchases, etc. in support of RAILS; and
- MRIM is the SPA.

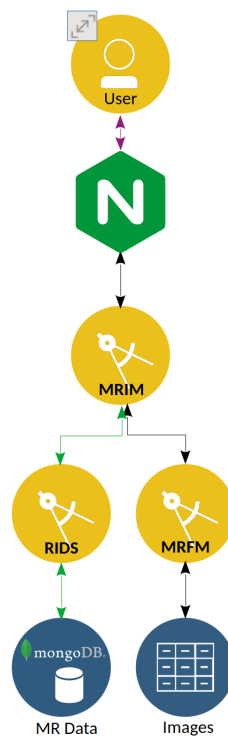


Figure 1.1: Microservices Components

1.3 System Components

Three system components that make up this subsystem:

- Network is a Transmission Control Protocol/Internet Protocol (TCP/IP) communication medium that connects the client (one or more) to the host running the MRIM docker containers;
- Server is a computer that runs MRIM microservices components; and
- Client¹ is a computer that runs a web browser to display the MRIM SPA.

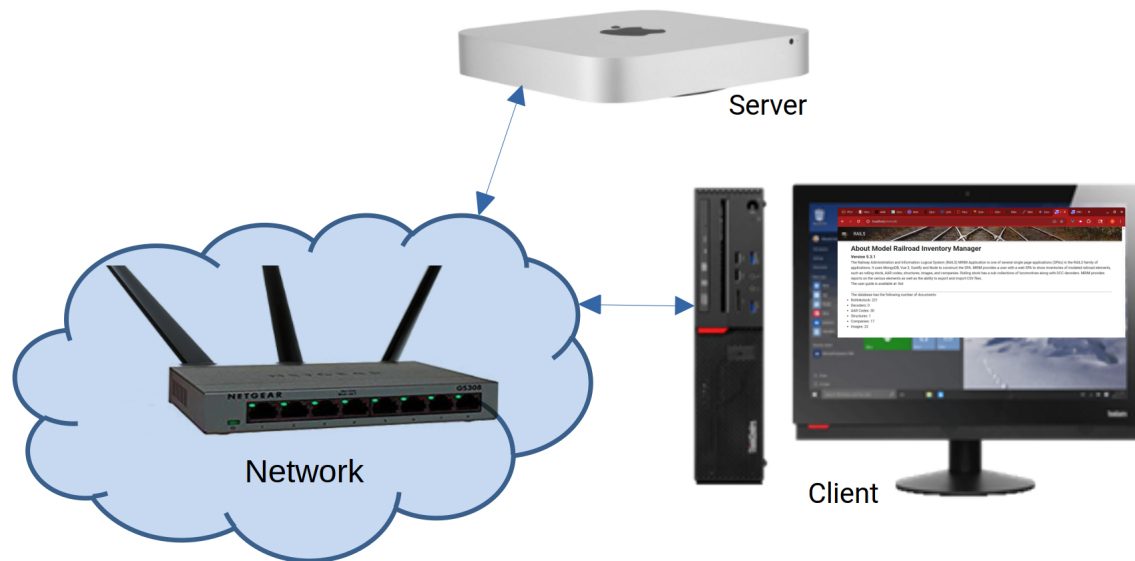


Figure 1.2: System Components

¹A web browser may be used on the Server and client machine is omitted

Chapter 2

Docker Set Up

2.1 Docker Installation

The RAILS SPAs are implemented as Docker containers. The Docker containers are built and pushed to Docker Hub. The Docker containers are then pulled from Docker Hub and run on the host machine. The host machine must have Docker installed. Docker is a platform for developing, shipping, and running applications in containers. Docker can be installed on Windows, macOS, and Linux. The installation instructions for Docker can be found at <https://docs.docker.com/get-docker/>.

2.2 Docker Compose Installation

Docker Compose is a tool for defining and running multi-container Docker applications. Docker Compose uses a yet another markup language (YAML) file to configure the application's services. The installation instructions¹ for Docker Compose can be found at <https://docs.docker.com/compose/install/>.

2.3 Broker Configuration

The RAILS SPAs use the Mosquitto broker to communicate with each other. The Mosquitto broker is an open-source message broker that implements the Message Queuing Telemetry Transport (MQTT) protocol. The Mosquitto broker must be configured to allow the RAILS SPAs to communicate with each other. The Mosquitto broker configuration file must be edited to allow the RAILS SPAs to communicate with each other. The configuration file is found in the virtual storage whose volume name is "mosquitto". To establish and modify the configuration file the following steps are taken:

1. Create a folder/directory "RAILS" and open a terminal from that folder/directory (all docker commands should be run in that terminal).
2. Create a Docker volume for the Mosquitto broker.

¹Note that for Microsoft Windows the installation of the Graphic User Interface (GUI) Docker Desktop automatically installs Docker Compose.

```
docker volume create --name mosquitto
```

3. Run a Docker container for the first time the Mosquitto broker, which will create the initial configuration file.

```
docker run -it --name myMqttBrkr -p 1883:1883 -p 9001:9001 --rm  
-v mosquitto:/mosquitto -d eclipse-mosquitto
```

4. Stop the broker container.

```
docker stop myMqttBrkr
```

5. Locate the configuration file in the "mosquitto" volume.

```
docker inspect mosquitto
```

6. Edit the configuration file modifying or adding the following lines:

```
listener 1883  
allow_anonymous true  
socket_domain ipv4
```

2.4 Create the RAILS Docker Environment

The **RAILS SPAs** are implemented as Docker containers that require an environment to run in. To create that environment, the following steps are taken:

1. Create a Docker volume for the Rails database.

```
docker volume create --name myRailsDb
```

2. Create a Docker volume for the Rails images.

```
docker volume create --name myRailsImages
```

3. Create a Docker network for the Rails containers to communicate over.

```
docker network create myRailsNet
```

2.5 Run the RAILS Docker Containers

The Docker containers are pulled from Docker Hub and run on the host machine. To run all of the RAILS Docker containers retrieve the YAML “docker-compose.yaml” file from the RAILS GitHub repository found at [my GitHub repository](#). Additionally retrieve the “nginx.conf” file from RAILS GitHub repository. Be sure that both files are put into the “RAILS” folder/directory. To fetch the Docker images, create the containers, and start the containers running execute the following command in the terminal opened in the “RAILS” folder/directory:

```
docker compose up -d
```

This command will pull the Docker images from Docker Hub and run the containers in detached mode. The RAILS SPAs will be running in the background, and the terminal will return to the command prompt. The RAILS SPAs will be running on the host machine. Point a browser, on the host machine, to the following URLs to access the RAILS SPAs:

- MRIM SPA - <http://localhost/mrim> (or http://<MACHINE_IP>/mrim/)
- RSRM SPA - <http://localhost/rsrm> (or http://<MACHINE_IP>/rsrm/)
- MPPM SPA - <http://localhost/mppm> (or http://<MACHINE_IP>/mppm/)
- MRLM SPA - <http://localhost/mrlm> (or http://<MACHINE_IP>/mrlm/)

Where <MACHINE_IP> is the IP address of the host machine running the Docker containers. The RAILS SPAs can be accessed from any browser on any Personal Computer (PC) in the network.

2.6 Stop and Remove the RAILS Docker Containers

To stop and remove the RAILS Docker containers, the following command is used:

```
docker compose down
```

Chapter 3

MRIM Operations

RSRM simplifies identifying rollingstock by associating Radio Frequency Identification (RFID) tags with specific road names and numbers. When a connected reader reads an RFID tag, RSRM searches its database for a matching entry. If a match is found, the application displays the corresponding road name and number of the rollingstock. If no match is found, the application provides input fields for the user to manually enter this information, creating a new association in the database.

3.1 About Page

The initial page when the application is opened in a web browser provides:

- A version number of the application;
- A concise overview of the application's purpose and technology stack;
- The current database statistics indicating:
 - The number of rollingstock entries in the database and how many have RFID tags; and
 - The number of micro-controllers in the database and how many are RFID readers.

Figure 3.1 shows the "About" page of the RAILS RS RFID Manager application.

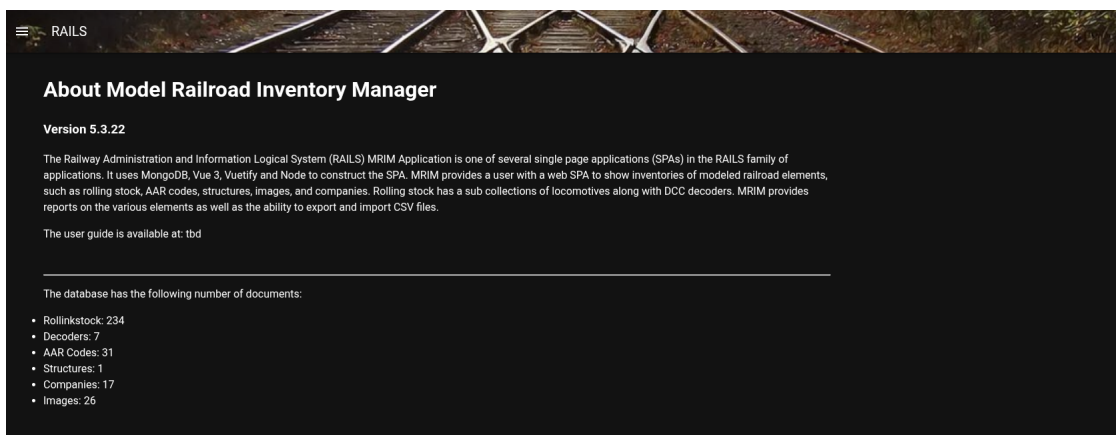


Figure 3.1: About Page

The hamburger menu icon ☰, to the left of "RAILS" represents a hidden navigation menu that can be toggled open and closed. When the icon is clicked or tapped on the menu for navigation to different pages of this application is revealed. This menu contains the links: "Reader," "Admin," "Rollingstock," "AAR Codes," "About" (this page). Figure 3.2 shows the "About" page with the navigation menu opened.

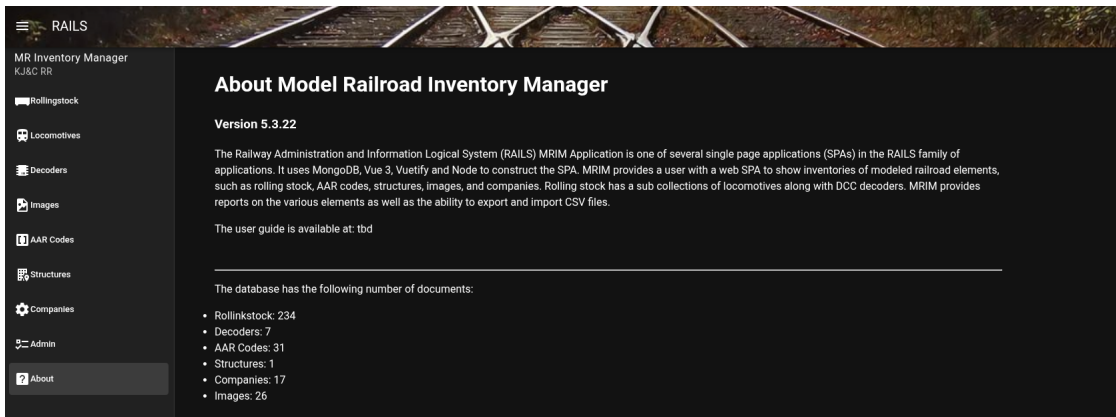


Figure 3.2: Sidemenu

3.2 Rolling Stock Management

The Rollingstock view provides a complete inventory of all rolling stock registered in the system. Selecting "Rollingstock" from the navigation drawer displays the main tabular view containing all registered pieces, including key attributes such as Road Name, Road Number, AAR Code, Description, Color, and available Action buttons. Figure 3.3 displays the main Rollingstock table page.

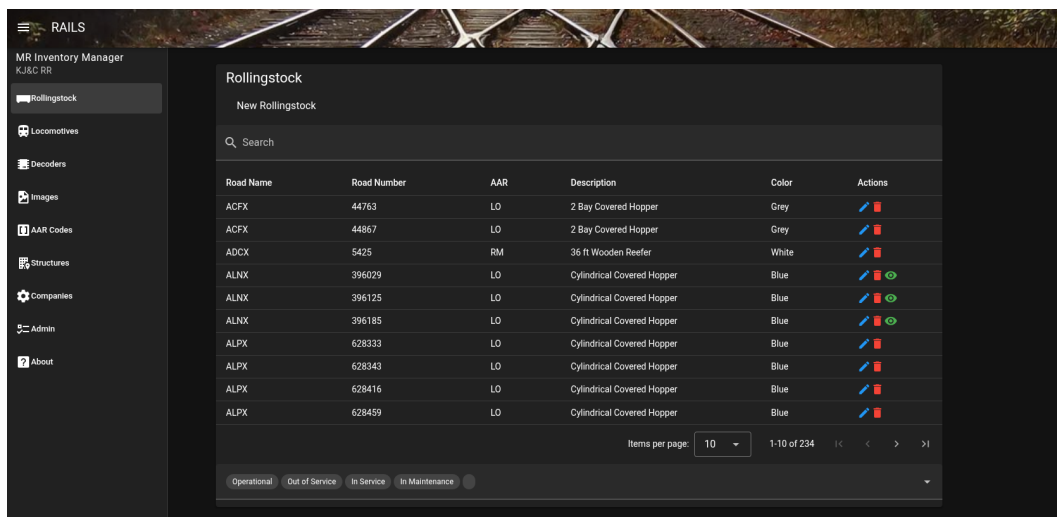


Figure 3.3: Rollingstock Page

3.2.1 Page Quantity Selection

The table displays items in paginated format. Users can control the number of rolling stock items shown per page using the dropdown selector located at the bottom right of the table. Available options include 10, 25, 50, 100, or All items. Figure 3.4 illustrates changing the items per page display count.

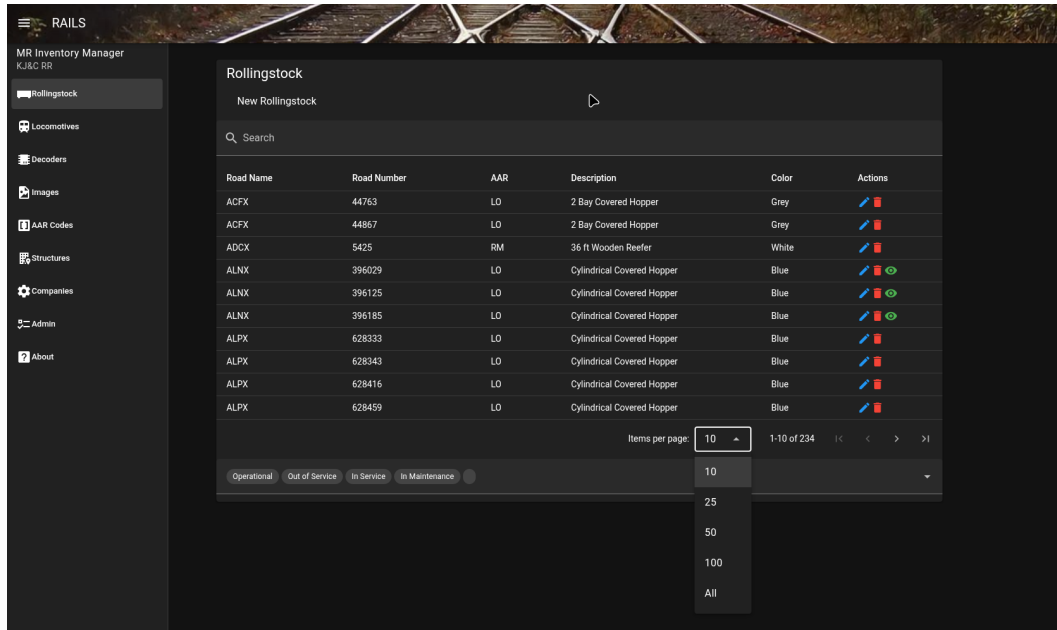


Figure 3.4: Items Per Page Selection

3.2.2 Status Filtering

Rolling stock entries can be filtered dynamically based on operational status. At the bottom of the view, status filter controls allow toggling visibility for items designated as Operational, Out of Service, In Service, or In Maintenance. Figure 3.5 shows the operational status filtering interface open.

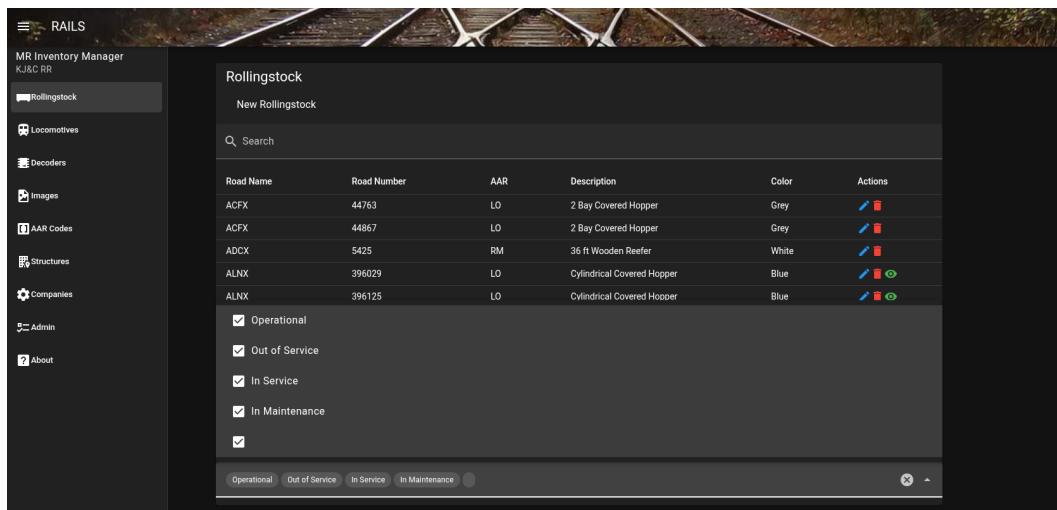


Figure 3.5: Status Filter

3.2.3 Search Functionality

A search bar located above the main table enables real-time filtering across rolling stock attributes. Typing search terms such as a road name (e.g., "cnw") instantly filters the list to show matching records. Figure 3.6 shows filtered search results.

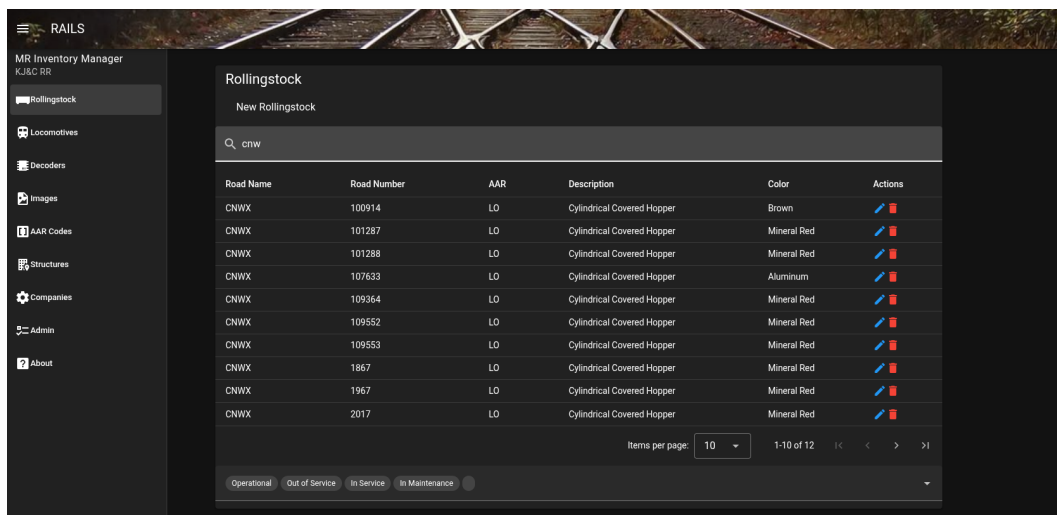


Figure 3.6: Search Rollingstock

3.2.4 Viewing Rollingstock Details

For entries configured with associated image files, an eye icon appears under the Actions column. Clicking this icon opens the Rollingstock Details dialog, displaying comprehensive prototype specifications, dimensions, status, and a photograph of the equipment. Figure 3.7 shows the detailed view modal for a selected car.

Rollingstock Details

Road Name: ALNX

Road Number: 396185

AAR Code: LO

Color: Blue

Description: Cylindrical Covered Hopper

Number Built: Builder: NSC Built Date: In Service Date: 1981-02-26

Inside Length: 59 Inside Height: Inside Width: Lt Weight: 61800

Load Limit: 201200 Load Types: Grain Capacity: 4550 Home Location:

Last Maintenance: 1975-06-29 Status: Operational

Model Details

RFID Tag: AAADCCDAED Weight: Length: Image ID: 45103wl.jpg

Notes:

Close

Description	Color	Actions
Cylindrical Covered Hopper	Blue	
Cylindrical Covered Hopper	Blue	
Cylindrical Covered Hopper	Blue	
Cylindrical Covered Hopper	Blue	
Cylindrical Covered Hopper	Blue	
Cylindrical Covered Hopper	Blue	
Cylindrical Covered Hopper	Blue	
Cylindrical Covered Hopper	Blue	
Cylindrical Covered Hopper	Blue	
Cylindrical Covered Hopper	Blue	

Items per page: 10 1-10 of 234

Figure 3.7: View Rollingstock Details

3.2.5 Adding New Rollingstock

Clicking the “New Rollingstock” button above the table opens a blank data entry form. Users can populate fields including road information, dimensions, load characteristics, model parameters, RFID tag associations, and linked image IDs. Figure 3.8 shows the entry form for creating a new rolling stock item.

New Rollingstock

Road Name: Road Number: AAR Code: Color:

Description:

Number Built: Builder: Built Date: In Service Date:

Inside Length: Inside Height: Inside Width: Lt Weight:

Load Limit: Load Types: Capacity: Home Location:

Last Maintenance: Status: ▼

Model Details

RFID Tag: Weight: Length: Image ID:

Notes:

Cancel Save

AAR	Description	Color	Actions
LO	2 Bay Covered Hopper	Grey	✎ ✖
LO	2 Bay Covered Hopper	Grey	✎ ✖
RM	36 ft Wooden Reefer	White	✎ ✖
LO	Cylindrical Covered Hopper	Blue	✎ ✖ ↻
LO	Cylindrical Covered Hopper	Blue	✎ ✖ ↻
LO	Cylindrical Covered Hopper	Blue	✎ ✖ ↻
LO	Cylindrical Covered Hopper	Blue	✎ ✖ ↻
LO	Cylindrical Covered Hopper	Blue	✎ ✖ ↻
LO	Cylindrical Covered Hopper	Blue	✎ ✖ ↻
LO	Cylindrical Covered Hopper	Blue	✎ ✖ ↻

Items per page: 10 1-10 of 234 ◀ > ▶

Figure 3.8: New Rollingstock Entry

3.2.6 Editing Rollingstock Records

To modify an existing entry, click the blue pencil icon in the Actions column for that row. This opens the details dialog in editable mode, populated with current record data for updating field values. Figure 3.9 illustrates editing an existing record.

Rollingstock Details

Road Name: ACFX Road Number: 44763 AAR Code: LO Color: Grey

Description: 2 Bay Covered Hopper

Number Built: Builder: HST Built Date: 1981-04-26 In Service Date:

Inside Length: 39 Inside Height: Inside Width: Lt Weight: 57200

Load Limit: 205800 Load Types: Capacity: 2980 Home Location: Ottawa

Last Maintenance: 1975-06-28 Status: Operational ▼

Model Details

RFID Tag: AAADCDDAE0 Weight: Length: Image ID:

Notes:

Cancel Save

AAR	Description	Color	Actions
LO	2 Bay Covered Hopper	Grey	✎ ✖
LO	2 Bay Covered Hopper	Grey	✎ ✖
RM	36 ft Wooden Reefer	White	✎ ✖
LO	Cylindrical Covered Hopper	Blue	✎ ✖ ↻
LO	Cylindrical Covered Hopper	Blue	✎ ✖ ↻
LO	Cylindrical Covered Hopper	Blue	✎ ✖ ↻
LO	Cylindrical Covered Hopper	Blue	✎ ✖ ↻
LO	Cylindrical Covered Hopper	Blue	✎ ✖ ↻
LO	Cylindrical Covered Hopper	Blue	✎ ✖ ↻
LO	Cylindrical Covered Hopper	Blue	✎ ✖ ↻

Items per page: 10 1-10 of 234 ◀ > ▶

Figure 3.9: Edit Rollingstock

3.2.7 Deleting Rollingstock Records

To remove a record from the database, click the red trash bin icon in the Actions column. A confirmation dialog appears requesting confirmation before permanently removing the record. Figure 3.10 shows the deletion confirmation prompt.

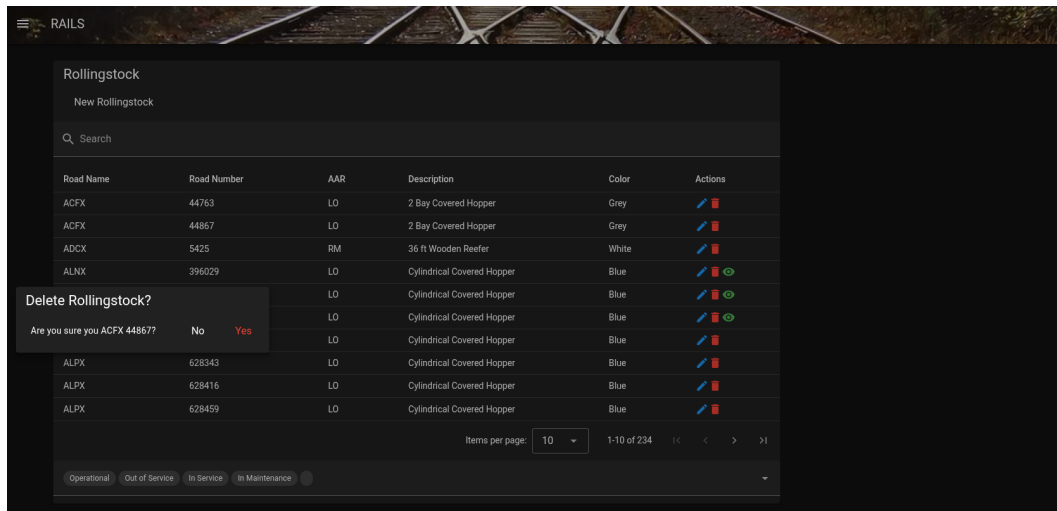


Figure 3.10: Delete Rollingstock Confirmation

3.3 Locomotives Inventory

Selecting "Locomotives" from the navigation drawer displays the Inventory of Locomotives view. This table presents a specialized subset of motive power entries from the inventory database, detailing primary attributes such as Road Name, Road Number, AAR designation code, Description, and Color scheme.

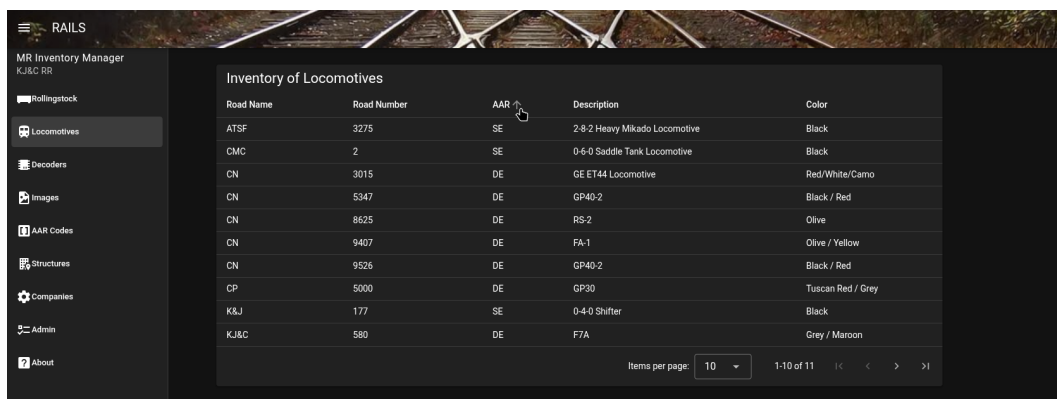


Figure 3.11: Inventory of Locomotives

The page incorporates interactive column sorting, as shown by the direction arrow next to the AAR header, allowing motive power records to be ordered by specific attributes. Standard table pagination controls located at the bottom right enable users to adjust the display limit per page and navigate through larger locomotive fleets.

3.4 Decoders Management

Selecting “Decoders” from the navigation drawer displays the main Decoders management view. This page displays a tabular listing of all Digital Command Control (DCC) decoders installed in inventory locomotives, summarizing information such as Road Name, Road Number, Manufacturer, Family, Model, active DCC Address, and available action controls. Figure 3.12 shows the main Decoders table page.

Road Name	Road Number	Manufacturer	Family	Model	Address	Actions
ATSF	3275	Broadway Limited	Paragon 3	Rolling Thunder	3275	[Edit] [Delete]
CMC	2	Bachmann Trains	E-Z Command decoders	4 function decoder (36-550)	2	[Edit] [Delete]
CN	3015	Electronic Solutions Ulm GmbH	LokSound	5	3015	[Edit] [Delete]
CN	5347	Digtrax	Economy Series 3 with FX3, silent	DH123 Four Function	5347	[Edit] [Delete]
KJ&C	580	Digtrax	Economy Series 3 with FX3, silent	DH123 Four Function	580	[Edit] [Delete]
VIA	6922	Electronic Solutions Ulm GmbH	LokSound	5	6922	[Edit] [Delete]
CN	8625	Digtrax	Economy Series 3 with FX3, silent	DH123 Four Function	8625	[Edit] [Delete]

Figure 3.12: Decoders Page

3.4.1 Adding a New Decoder

Clicking the “Add Decoder” button above the table opens a modal dialog for registering a new decoder entry. The form includes fields for Road Name, Road Number, Manufacturer, Family, Model, and DCC Address. Figure 3.13 shows the new decoder entry dialog.

New Decoder

Road Name

Road Number

Manufacturer

Family

Model

Address

Cancel Save

Figure 3.13: Add Decoder

3.4.2 Editing Decoder Records

To modify an existing decoder entry, click the blue pencil icon under the Actions column. This opens the decoder configuration form pre-populated with current settings, allowing updates to hardware details or DCC address assignments. Figure 3.14 displays the edit decoder interface.

Family	Model	Address	Actions
Paragon 3	Rolling Thunder	3275	
E-Z Command decoders	4 function decoder (36-550)	2	
LokSound	5	3015	
Economy Series 3 with FX3, silent	DH123 Four Function	5347	
Economy Series 3 with FX3, silent	DH123 Four Function	580	
LokSound	5	6922	
Economy Series 3 with FX3, silent	DH123 Four Function	8625	

Items per page: 10 1-7 of 7

Figure 3.14: Edit Decoder

3.4.3 Deleting Decoder Records

To delete a decoder entry, click the red trash bin icon in the Actions column. A modal confirmation prompt appears requesting verification before the record is permanently removed from the database. Figure 3.15 illustrates the deletion confirmation dialog.

Road Name	Road Number	Manufacturer	Family	Model	Address	Actions
ATSF	3275	Broadway Limited	Paragon 3	Rolling Thunder	3275	
CMC	2	Bachmann Trains	E-Z Command decoders	4 function decoder (36-550)	2	
CN	3015	Electronic Solutions Ulm GmbH	LokSound	5	3015	
CN	5347	Digitrax	Economy Series 3 with FX3, silent	DH123 Four Function	5347	
K/J&C	580	Digitrax	Economy Series 3 with FX3, silent	DH123 Four Function	580	
VIA	6922	Electronic Solutions Ulm GmbH	LokSound	5	6922	
	8625	Digitrax	Economy Series 3 with FX3, silent	DH123 Four Function	8625	

Items per page: 10 1-7 of 7

Figure 3.15: Delete Decoder Confirmation

3.5 Image Management

Selecting "Images" from the navigation drawer displays the Images management view. This page displays a tabular repository of all uploaded graphic assets stored in the Model Railroad File

Manager (MRFM) microservice images, detailing each item's Title, File Name, assigned Category (e.g., Rollingstock or Structure), and Action buttons. Figure 3.16 shows the main Images view.

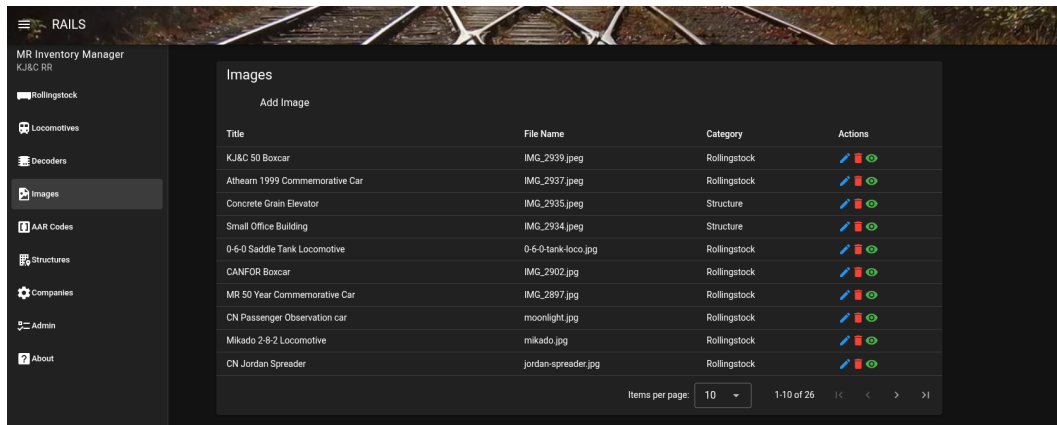


Figure 3.16: Images Page

3.5.1 Viewing Image Details

Clicking the green eye icon under the Actions column opens a modal dialog showing full meta-data and a visual preview of the selected asset. The dialog presents the Title, File Name, Category, optional Notes, and an embedded image preview. Figure 3.17 shows the image viewer dialog.

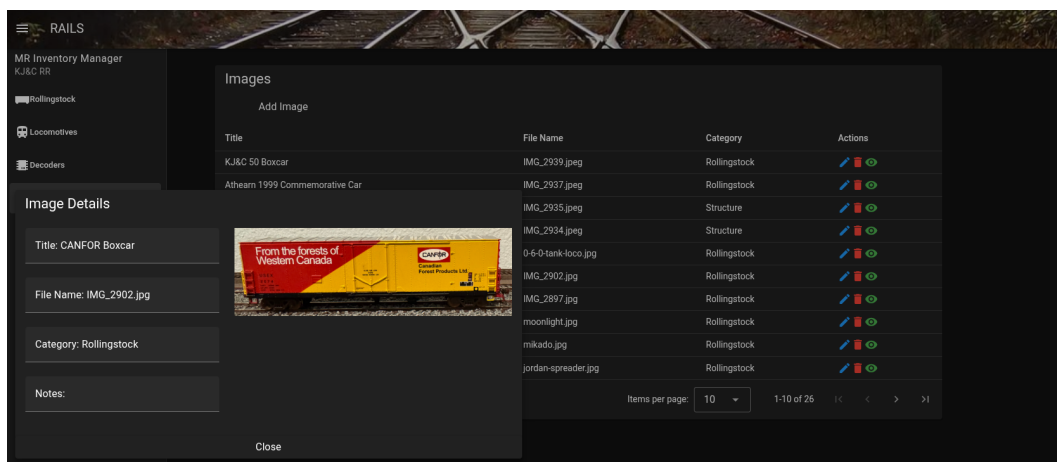


Figure 3.17: View Image Details

3.5.2 Adding a New Image

Clicking the "Add Image" button above the table opens a modal form for registering and uploading new graphic files. The form prompts for a Title, a local File input path, a Category assignment, and optional Notes. Figure 3.18 illustrates the upload modal.

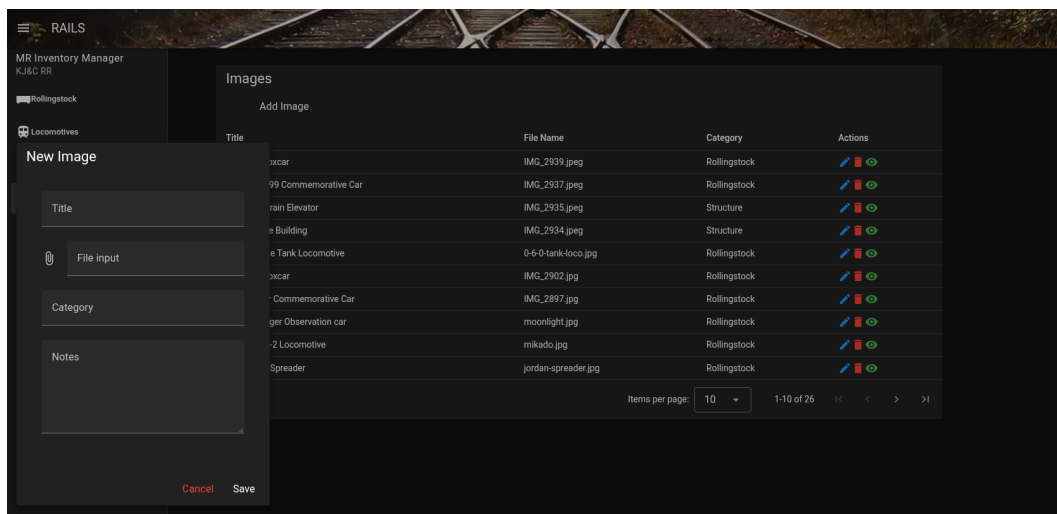


Figure 3.18: Add Image Dialog

3.5.3 Editing Image Metadata

To modify an existing image entry's descriptive metadata, click the blue pencil icon in the Actions columnimages. This opens the image dialog pre-populated with current values, allowing edits to the Title, File Name, Category, and Notes. Figure 3.19 shows the edit dialog.

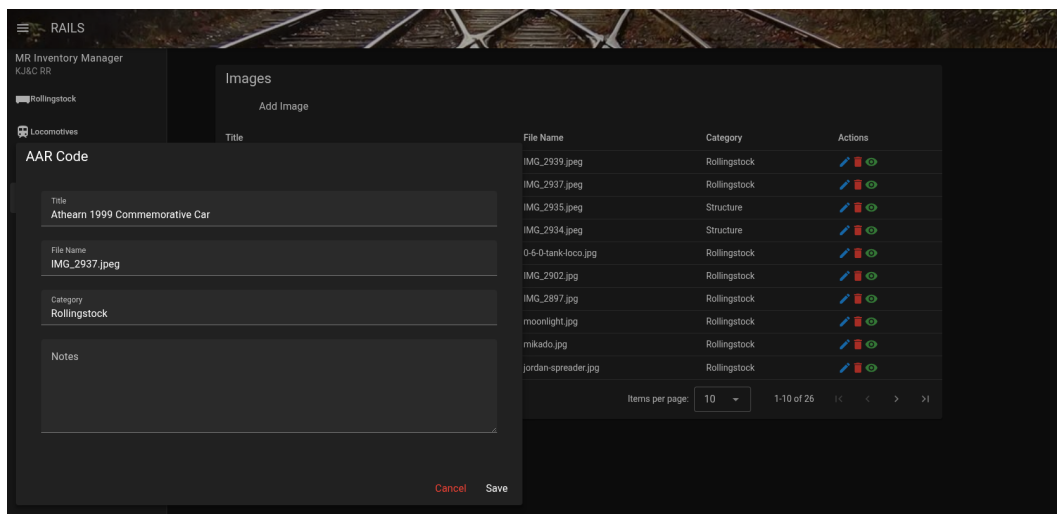


Figure 3.19: Edit Image Metadata

3.5.4 Deleting Image Records

To remove an image from the database repository, click the red trash bin icon under the Actions columnimages. A modal confirmation prompt appears requesting verification before the image asset record is deleted. Figure 3.20 displays the deletion confirmation prompt.

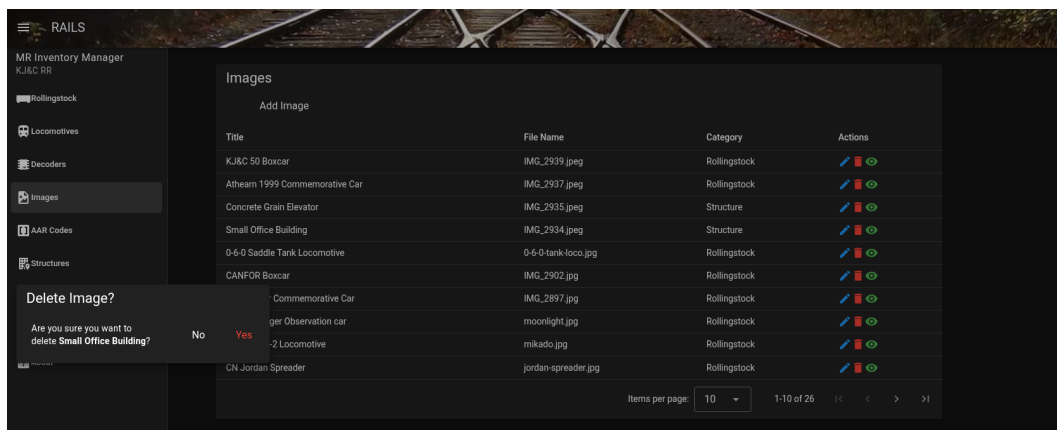


Figure 3.20: Delete Image Confirmation

3.6 AAR Code Reference

Selecting "AAR Codes" from the navigation drawer opens the Association of American Railroads (AAR) designation reference view. This page provides a standardized lookup table for rolling stock classification codes used throughout the inventory system, displaying the AAR Code, Rollingstock Type (RS Type), standard Description, and Action controls. Figure 3.21 illustrates the main AAR Codes reference table.

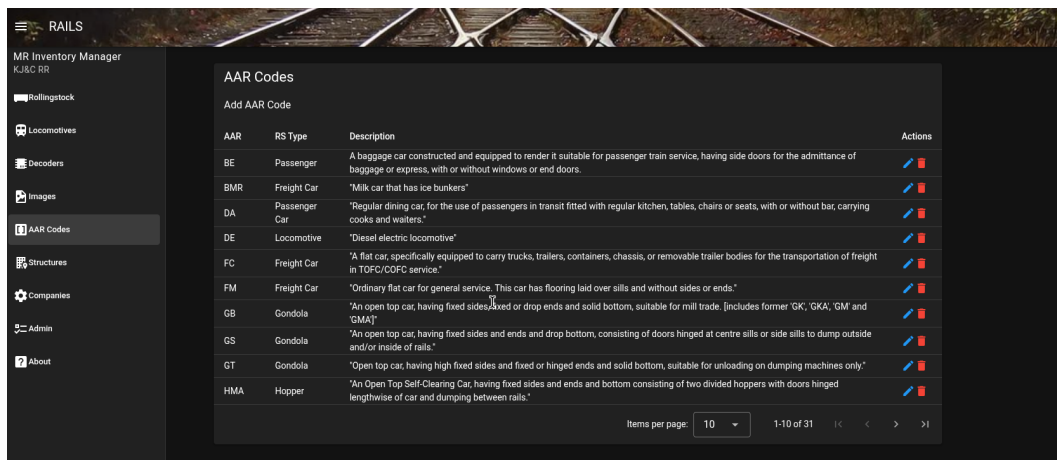


Figure 3.21: AAR Codes Page

3.6.1 Adding an AAR Code

Clicking the "Add AAR Code" button above the table opens a modal dialog to define a new equipment classification code. The form requires an AAR Code abbreviation, the associated RS Type, and a comprehensive Description detailing prototype car characteristics. Figure 3.22 shows the entry modal.

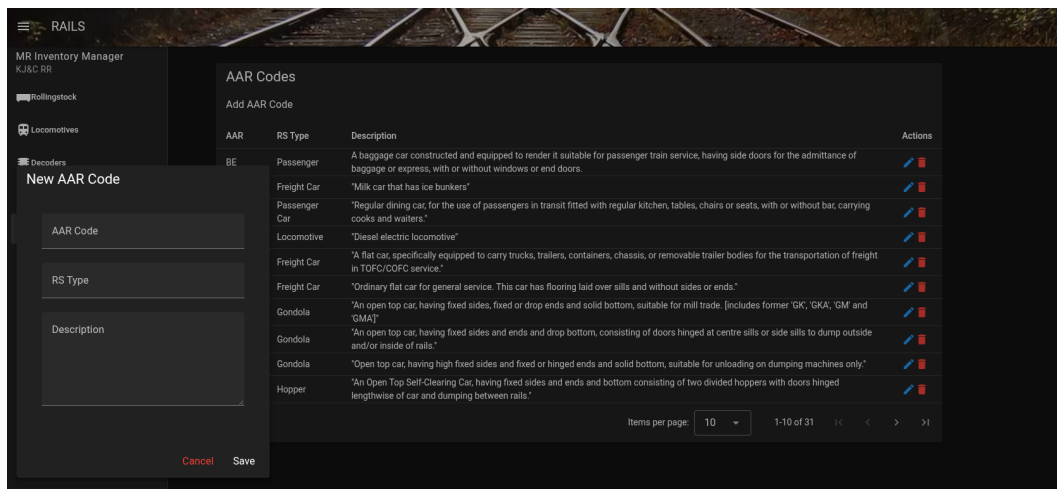


Figure 3.22: Add AAR Code Dialog

3.6.2 Editing AAR Codes

To modify an existing classification entry or update its prototype definition, click the blue pencil icon under the Actions column. This displays the AAR Code form with populated values for editing. Figure 3.23 shows the edit dialog interface.

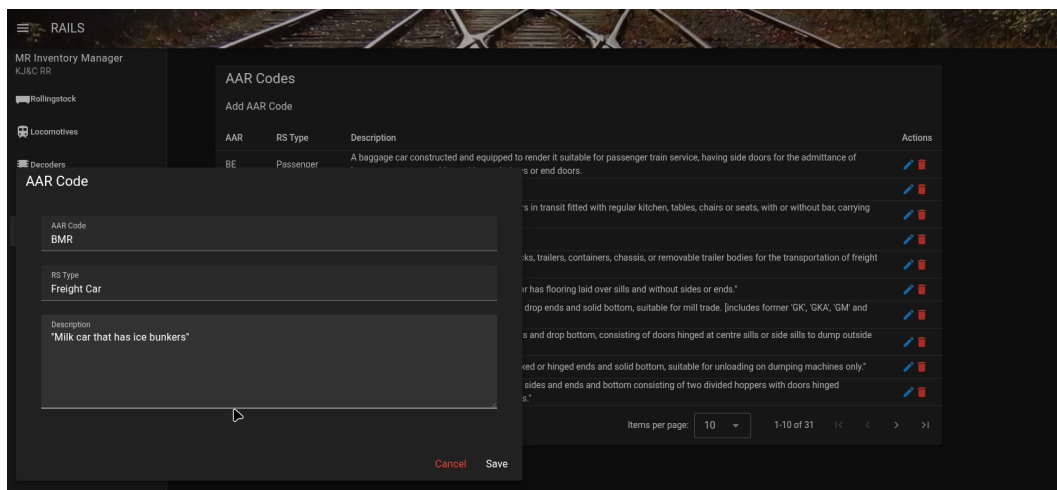


Figure 3.23: Edit AAR Code

3.6.3 Deleting AAR Codes

To remove a classification code from the system lookup table, click the red trash bin icon in the Actions column. A confirmation modal prompt appears requiring verification before the AAR code entry is permanently deleted. Figure 3.24 shows the deletion confirmation prompt.

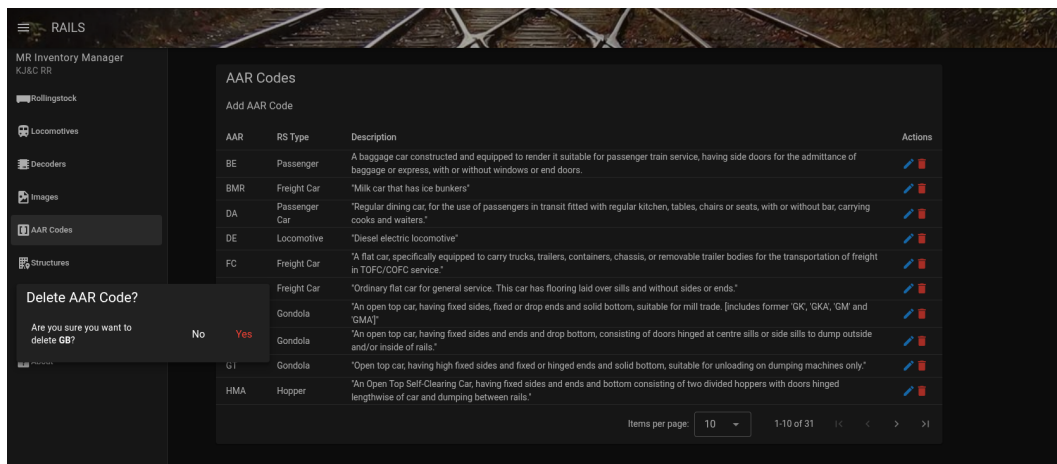


Figure 3.24: Delete AAR Code Confirmation

3.7 Structures Management

Selecting "Structures" from the navigation drawer displays the Structures management view. This page presents a tabular inventory of layout buildings and trackside structures, summarizing attributes such as Title, Use, Owner, layout Location, Year Built, and Action controls. Figure 3.25 shows the main Structures table page.

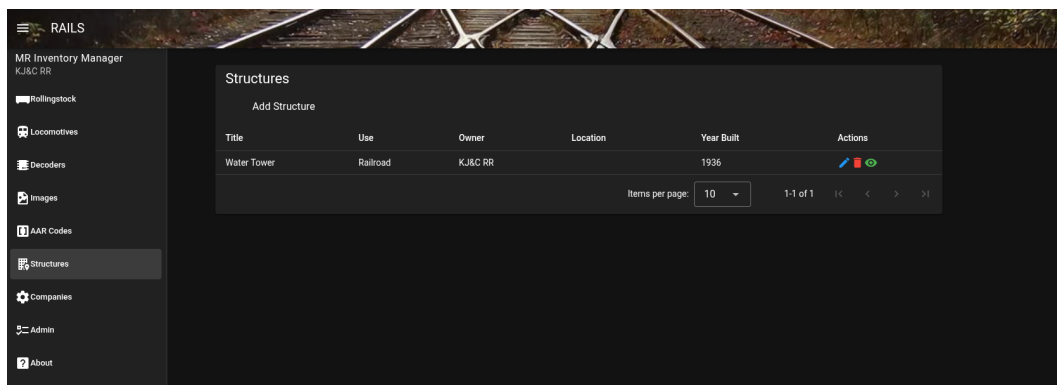


Figure 3.25: Structures Page

3.7.1 Viewing Structure Details

Clicking the green eye icon in the Actions column opens a detail view modal. This dialog presents recorded attributes alongside an embedded photo preview of the modeled structure. Figure 3.26 displays the structure inspection dialog.

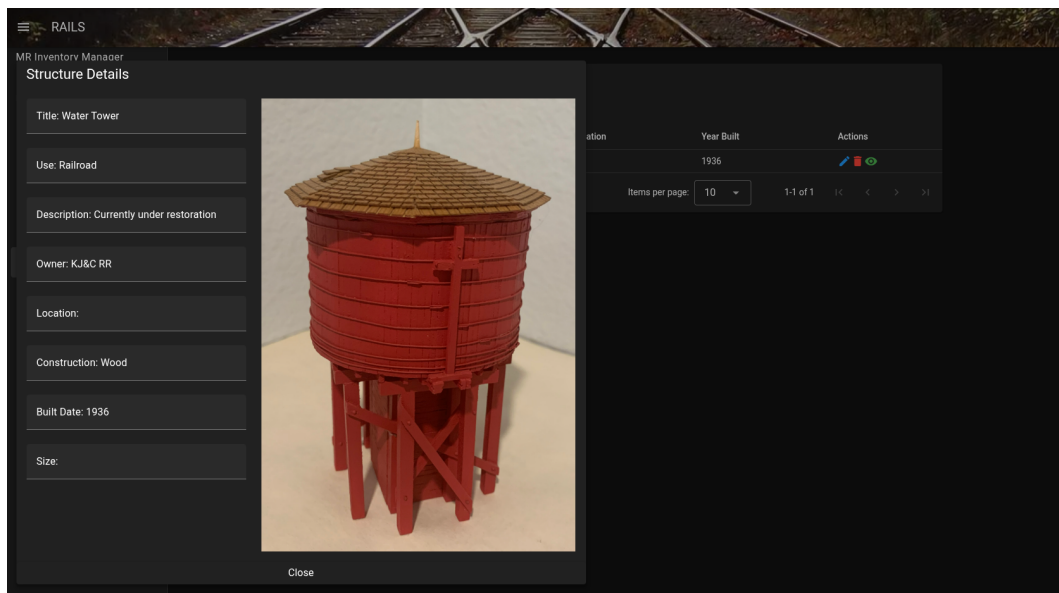


Figure 3.26: View Structure Details

3.7.2 Adding a New Structure

Clicking the “Add Structure” button above the table opens a modal dialog for adding a new building or facility to the database. The form includes comprehensive fields for general metadata (Title, Use, Owner, Location, Construction, Built Date, Size, Image) and operational parameters (Industrial classification, Type, Raw Materials, Capacities, Production Rates, Priority, AAR Codes In/Out, Load/Unload Durations, Siding Capacity, and Notes). Figure 3.27 illustrates the structure creation modal.

New Structure

Title

Use

Description

Owner

Location

Construction

Built Date

Size

Image

Industrial
false

Type

Raw Materials

rMCapacity

conRate

Priority

AAR Code In

Product

Product Capacity

Product Rate

AAR Code Out

Unload Duration

Load Duration

Siding Capacity

notes

Cancel Save

Location	Year Built	Actions
	1936	  

Items per page: 10 1-1 of 1

Figure 3.27: Add Structure Dialog

3.7.3 Editing Structure Records

To update an existing entry, click the blue pencil icon under the Actions column. This opens the structure form populated with current records, allowing edits to physical attributes, image references, or industrial operational data. Figure 3.28 shows the edit structure dialog.

Structure

Title	Water Tower	Use	Railroad	Owner	KJ&C RR
Description	Currently under restoration		Location		
Construction	Wood	Built Date	1936	Size	
Image	IMG_1430.jpg	Industrial	false	Type	
Raw Materials		rMCapacity		conRate	
Priority		AAR Code In		Product	
Product Capacity		Product Rate		AAR Code Out	
Unload Duration		Load Duration		Siding Capacity	
Notes					

Cancel Save

Figure 3.28: Edit Structure

3.7.4 Deleting Structure Records

To remove a structure entry from the inventory, click the red trash bin icon in the Actions column. A confirmation prompt appears requiring verification before the record is deleted from the system database. Figure 3.29 illustrates the deletion modal dialog.

RAILS

MR Inventory Manager
KJ&C RR

Rollingstock
Locomotives
Decoders
Images
AAR Codes
Structures

Structures

Add Structure

Title	Use	Owner	Location	Year Built	Actions
Water Tower	Railroad	KJ&C RR		1936	  

Items per page: 10 1-1 of 1

Delete Structure?

Are you sure you want to delete Water Tower?

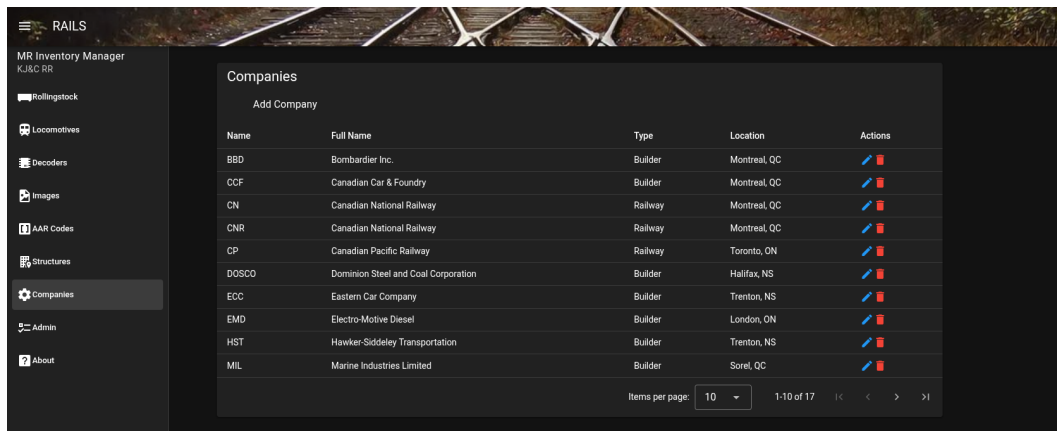
No Yes

Figure 3.29: Delete Structure Confirmation

3.8 Companies Reference

Selecting "Companies" from the navigation drawer opens the Companies management view. This page serves as a central reference table for railroad equipment manufacturers, locomotive builders, and operating railway companies. It displays the company short Name (abbreviation), Full Name,

Type classification (e.g., Builder, Railway), Location, and Action controls. Figure 3.30 illustrates the main Companies page interface.

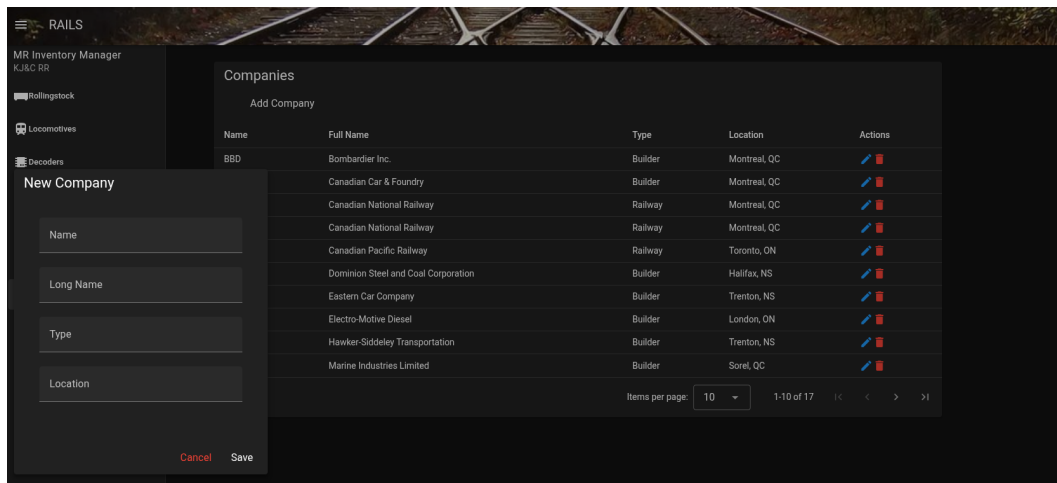


Name	Full Name	Type	Location	Actions
BBD	Bombardier Inc.	Builder	Montreal, QC	 
CCF	Canadian Car & Foundry	Builder	Montreal, QC	 
CN	Canadian National Railway	Railway	Montreal, QC	 
CNR	Canadian National Railway	Railway	Montreal, QC	 
CP	Canadian Pacific Railway	Railway	Toronto, ON	 
DOSCO	Dominion Steel and Coal Corporation	Builder	Halifax, NS	 
ECC	Eastern Car Company	Builder	Trenton, NS	 
EMD	Electro-Motive Diesel	Builder	London, ON	 
HST	Hawker-Siddeley Transportation	Builder	Trenton, NS	 
MIL	Marine Industries Limited	Builder	Sorel, QC	 

Figure 3.30: Companies Page

3.8.1 Adding a Company

Clicking the “Add Company” button above the table opens a modal dialog to define a new corporate entry. The form prompts for the company’s short Name, Long Name, Type (such as Builder or Railway), and headquarters Location. Figure 3.31 shows the company entry modal.



Name	Full Name	Type	Location	Actions
BBD	Bombardier Inc.	Builder	Montreal, QC	 
CCF	Canadian Car & Foundry	Builder	Montreal, QC	 
CN	Canadian National Railway	Railway	Montreal, QC	 
CNR	Canadian National Railway	Railway	Montreal, QC	 
CP	Canadian Pacific Railway	Railway	Toronto, ON	 
DOSCO	Dominion Steel and Coal Corporation	Builder	Halifax, NS	 
ECC	Eastern Car Company	Builder	Trenton, NS	 
EMD	Electro-Motive Diesel	Builder	London, ON	 
HST	Hawker-Siddeley Transportation	Builder	Trenton, NS	 
MIL	Marine Industries Limited	Builder	Sorel, QC	 

Figure 3.31: Add Company Dialog

3.8.2 Editing Company Records

To modify an existing company entry, click the blue pencil icon under the Actions column. This opens the edit dialog populated with current information, allowing adjustments to reporting marks, full corporate titles, or location details. Figure 3.32 displays the edit company interface.

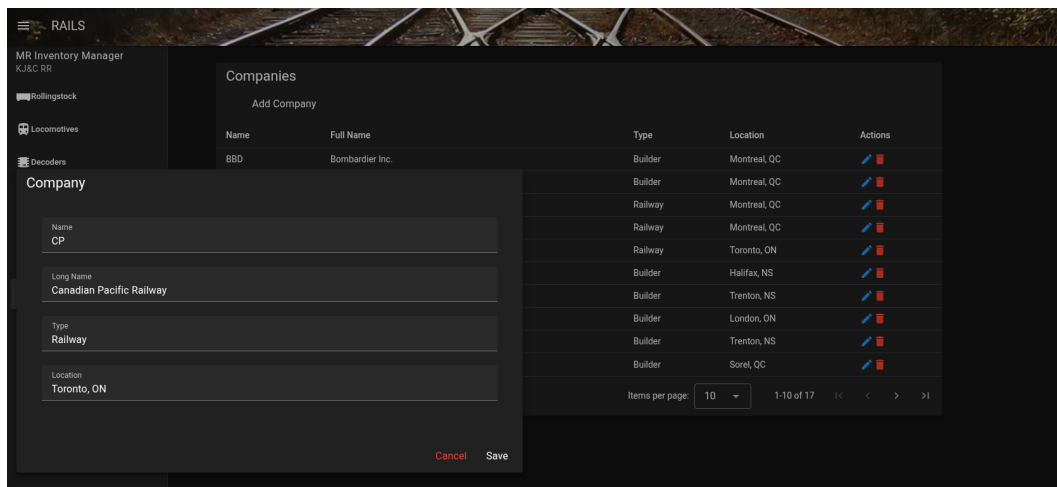


Figure 3.32: Edit Company

3.8.3 Deleting Company Records

To remove a corporate entity from the system reference table, click the red trash bin icon in the Actions column. A confirmation prompt appears asking for verification before removing the company entry. Figure 3.33 shows the deletion confirmation prompt dialog.

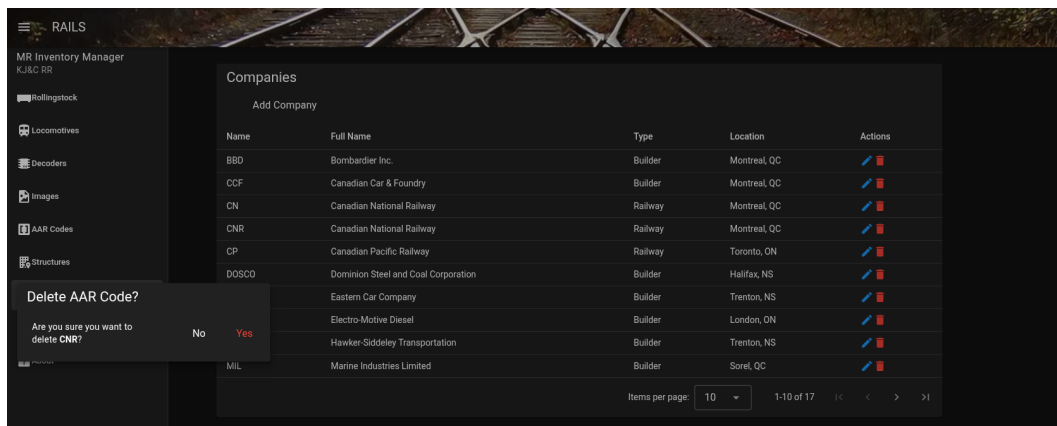


Figure 3.33: Delete Company Confirmation

3.9 Admin Dashboard

Selecting "Admin" from the navigation drawer opens the Admin dashboard view. This page provides centralized controls for system data management, including database collection exports, imports, report generation, and car card printing. Figure 3.34 shows the main Admin page interface.

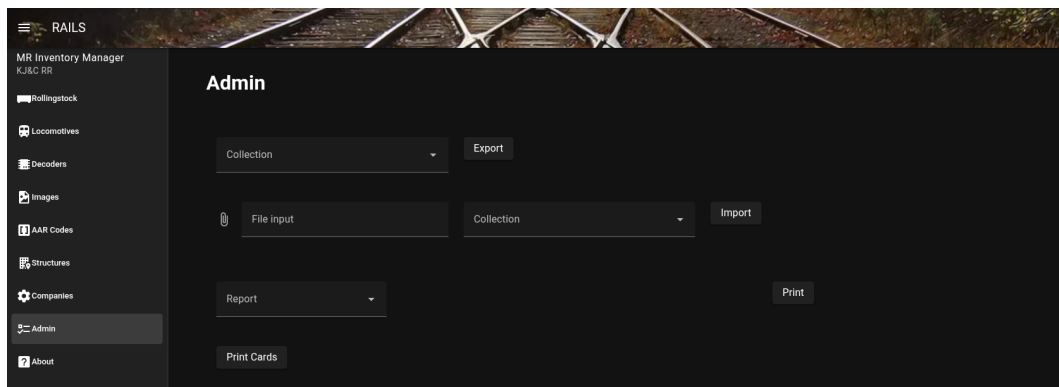


Figure 3.34: Admin Page

3.9.1 Exporting Data Collections

The Export control allows backup or data extraction of individual system collections. Selecting the "Collection" dropdown opens a list of available data sets, including AAR Codes, Companies, Images, Rolling Stock, Decoders, and Structures. Figure 3.35 illustrates the collection selection menu for data export.

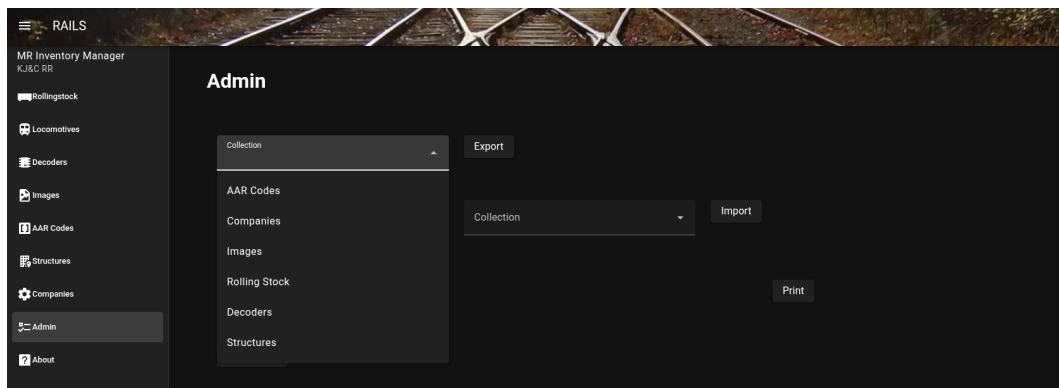


Figure 3.35: Select Export Collection

3.9.2 Importing Data Collections

To restore or bulk-load records, click the paperclip icon or file input field to launch the system file browser and select a local source file. Figure 3.36 shows the native file chooser selection dialog.

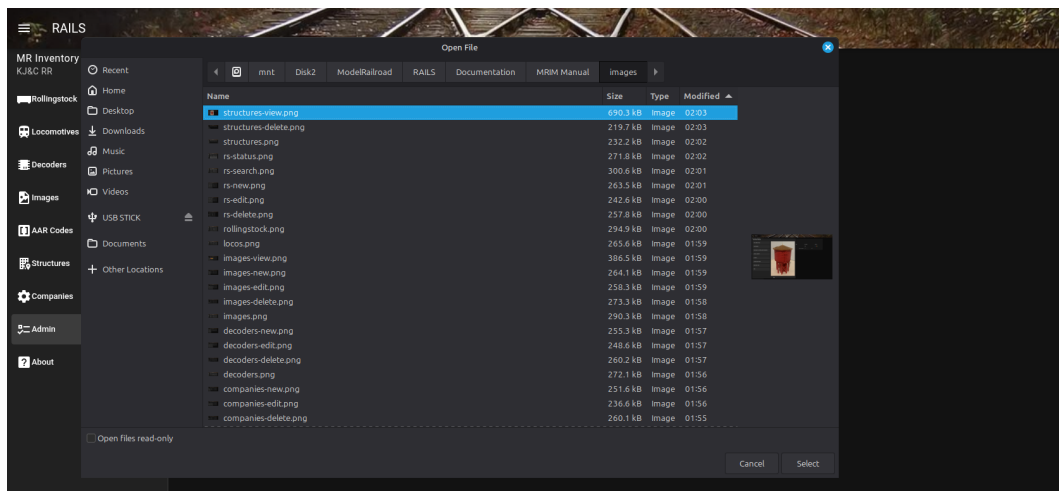


Figure 3.36: Select Import Source File

After specifying a source file, select the target database destination using the “Collection” dropdown menu. Options include AAR Codes, Companies, Images, Rolling Stock, Decoders, and Structures. Figure 3.37 illustrates selecting the target import collection before clicking “Import”.

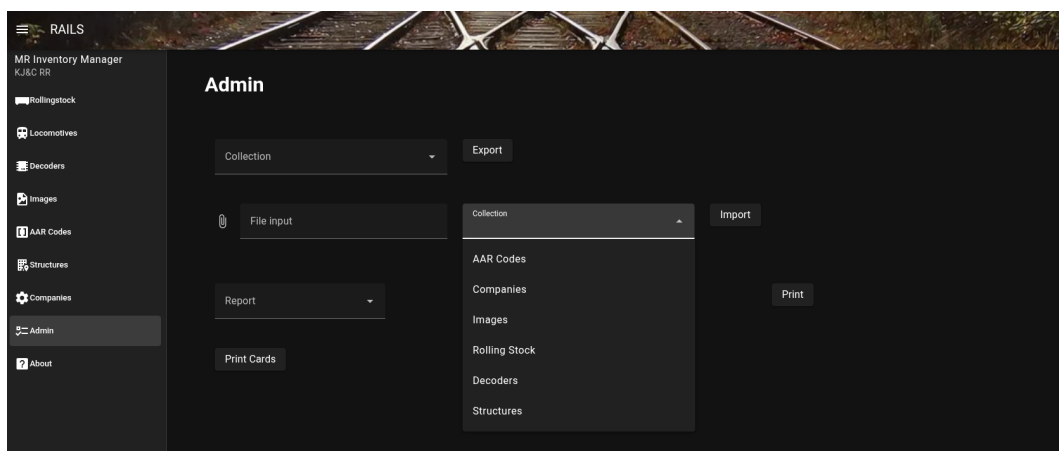


Figure 3.37: Select Target Import Collection

3.9.3 Generating System Reports

The Report section generates formatted system documentation and operational materials as a Portable Document Format (PDF) format. Selecting the “Report” dropdown displays available system inventories and assets for export, such as Rolling Stock, AAR Codes, Companies, Images, RFID, and Structures. Figure 3.38 shows the report type selection menu.

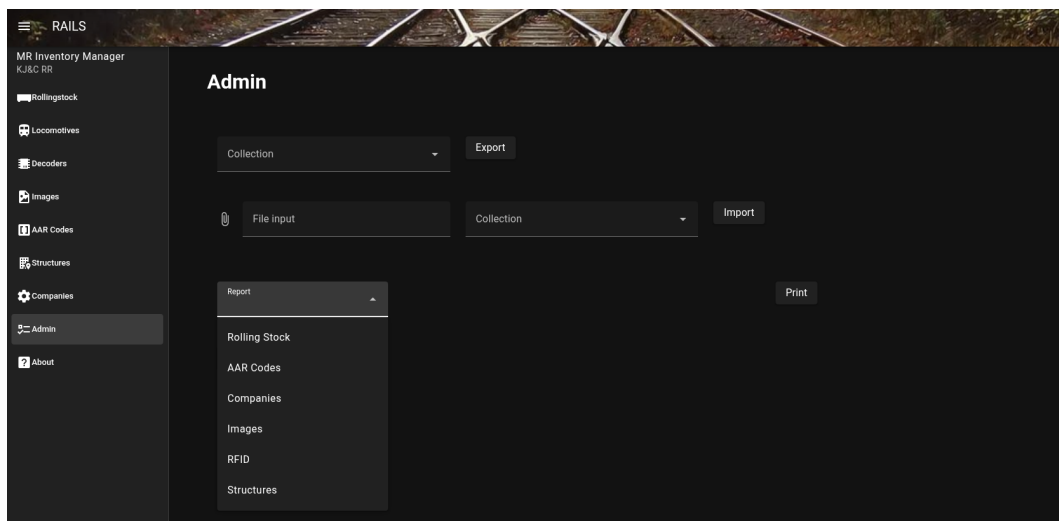


Figure 3.38: Select Report Category

When a specific report category like Rolling Stock is selected, additional sorting options appear. Clicking the "Sorted By" dropdown allows records to be ordered by parameters such as Road Names, AAR Codes, or Status. Figure 3.39 illustrates the sorting criteria options.

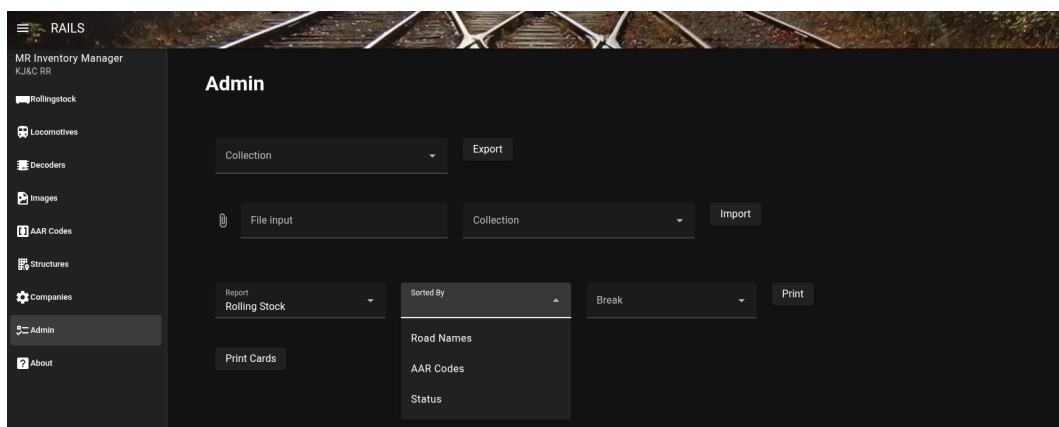


Figure 3.39: Report Sort Options

Reports can also be structured using page breaks or formatting style options. Selecting the "Break" dropdown provides options for Continuous formatting, Table views, or explicit Page breaks between sections. Figure 3.40 shows the report page break selection menu.

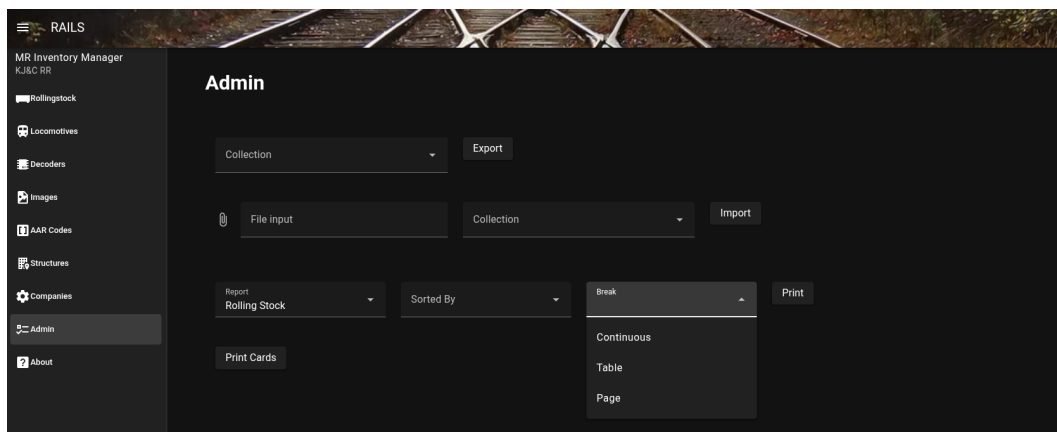


Figure 3.40: Report Page Break Formatting

Clicking on the "Print Cards" creates a set of pages that can be cut into car cards used as a two-part system to track freight cars and simulate real-world cargo movements across a layout.

Acronyms

Graphic User Interface (GUI) is the visual way you interact with a computer program using elements like windows, icons, menus, and buttons, instead of typing commands. GUIs are the standard way most people interact with computers today 6

Model Projects and Purchase Manager (MPPM) 3, 8

Message Queuing Telemetry Transport (MQTT) is a lightweight messaging protocol designed for machine-to-machine (M2M) communication in resource-constrained environments, like those found in the Internet of Things (IoT) 6

Model Railroad (MR) 4

Model Railroad File Manager (MRFM) 4

Model Railroad Inventory Manager (MRIM) 3, 4, 5, 8

Model Railroad Layout Manager (MRLM) 3, 8

Portable Document Format (PDF) 29

Railway Administration and Information Logical System (RAILS) 3, 4, 6, 7, 8

Representational State Transfer (REST) is a set of architectural principles for designing web services. It's not a protocol itself (like HTTP), but rather a guideline for creating web services 4

Radio Frequency Identification (RFID) is technology that uses radio waves to wirelessly identify and track objects 9

Railroad Inventory Data Services (RIDS) 4

rollingstock (RS) all the vehicles that operate on a railroad line. This includes both powered and unpowered vehicles 4

Rollingstock RFID Manager (RSRM) 3, 8, 9

Single Page Application (SPA) a web application that loads a single HTML page in the user's browser and dynamically updates the content based on user interaction, without reloading the entire page. This creates a more fluid and responsive user experience, similar to what you might expect from a native mobile app 3, 4, 5, 6, 7, 8

Transmission Control Protocol/Internet Protocol (TCP/IP) a suite of communication protocols that governs how data is exchanged over the internet or a private network. It is the foundation of any data exchange on the Web 5

yet another markup language (YAML) is a human-readable data serialization language used for configuration files, data exchange, and system automation, known for its use of indentation to define structure, similar to Python, and for being a strict superset of JSON 6, 8